

In-class exercise: code smells — solutions

Each exercise: **problem** slide → **discussion** (smells named) → **solution**. ***

Exercise 1

```
def f(x, s, n, k, a, b, flag):  
    from math import sqrt, pi, exp, comb  
    if flag == "normal":  
        return (1/(sqrt(2*pi)*s)) * exp(-0.5*((x-a)/s)**2)  
    elif flag == "binomial":  
        return comb(n, k) * (x**k) * ((1-x)**(n-k))  
    elif flag == "beta":  
        from math import gamma  
        return (x**(a-1)*(1-x)**(b-1))/(gamma(a)*gamma(b)/gamma(a+b))
```

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Exercise 1 — discussion

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-  Shotgun surgery risk — adding a distribution means editing this function.

Refactoring direction: Separate functions per distribution, each with descriptive parameter names.

Exercise 1 — one possible solution

```
from math import sqrt, pi, exp, comb, gamma

def normal_pdf(x, mean, sd):
    return (1/(sqrt(2*pi)*sd)) * exp(-0.5*((x-mean)/sd)**2)

def binomial_pmf(prob, n, k):
    return comb(n, k) * (prob**k) * ((1-prob)**(n-k))

def beta_pdf(x, alpha, beta_):
    B = gamma(alpha)*gamma(beta_)/gamma(alpha+beta_)
    return (x**(alpha-1) * (1-x)**(beta_-1)) / B
```

Exercise 2

```
def run(data, conds, sbjs, dprimes, criteria, mdl, fit, r, p, out):  
    r = []  
    for s in sbjs:  
        sd = [data[i] for i in range(len(data)) if conds[i] == s]  
        dp = sum([x[0] for x in sd]) / len(sd)  
        cr = sum([x[1] for x in sd]) / len(sd)  
        dprimes.append(dp)  
        criteria.append(cr)  
    fit = mdl(dprimes, criteria)  
    out = "significant" if fit.pvalue < 0.05 else "not significant"  
    return out
```

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Refactoring direction: Extract a helper for per-subject means; trim the parameter list. ***

Exercise 2 — one possible solution (part 1)

```
def subject_means(data, conditions, subject):  
    rows = [data[i] for i in range(len(data)) if conditions[i] == subject]  
    dprime = sum(r[0] for r in rows) / len(rows)  
    criterion = sum(r[1] for r in rows) / len(rows)  
    return dprime, criterion
```

Exercise 2 — one possible solution (part 2)

```
def fit_model(data, conditions, subjects, model):  
    dprimes, criteria = [], []  
    for s in subjects:  
        dp, cr = subject_means(data, conditions, s)  
        dprimes.append(dp)  
        criteria.append(cr)  
    return model(dprimes, criteria)
```

Exercise 2 — one possible solution (part 2)

```
def report_result(fit):  
    # ... check that fit is the right kind of object  
    if fit.pvalue < 0.05  
        return "significant"  
    if fit.pvalue >= 0.05  
        return "not significant"  
    # ... this should not happen, maybe throw an error here
```

```
def c(h, f):  
    from scipy.stats import norm  
    return norm.ppf(h) - norm.ppf(f)  
  
def b(h, f):  
    from scipy.stats import norm  
    return -0.5 * (norm.ppf(h) + norm.ppf(f))
```

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-  Duplicated code — `norm` is imported twice inside identical boilerplate.

Refactoring direction: Use descriptive names; hoist the shared import. ***

Exercise 3 — one possible solution

```
from scipy.stats import norm

def dprime(hit_rate, false_alarm_rate):
    return norm.ppf(hit_rate) - norm.ppf(false_alarm_rate)

def criterion(hit_rate, false_alarm_rate):
    return -0.5 * (norm.ppf(hit_rate) + norm.ppf(false_alarm_rate))
```

Exercise 4

```
def compute_sensitivity_index_for_signal_detection_analysis(hr, far):  
    from scipy.stats import norm  
    # Compute d-prime using the standard SDT formula  
    # hr is hit rate, far is false alarm rate  
    z_hit = norm.ppf(hr)    # z-score for hit rate  
    z_far = norm.ppf(far)   # z-score for false alarm rate  
    dp = z_hit - z_far     # d-prime is the difference  
    return dp              # return the result
```

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Refactoring direction: Shorter name; one docstring for the interface; no inline commentary. ***

Exercise 4 — one possible solution

```
from scipy.stats import norm

def dprime(hit_rate, false_alarm_rate):
    """Return d-prime (SDT sensitivity index)."""
    return norm.ppf(hit_rate) - norm.ppf(false_alarm_rate)
```

Exercise 5

```
import numpy as np

def analyze(results):
    results = [r for r in results if r is not None]
    results = [r * 1000 for r in results]
    results = np.array(results)
    results = results[results < np.percentile(results, 95)]
    results = results - results.mean()
    return results
```

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Refactoring direction: Give each transformation stage a distinct, descriptive name. ***

Exercise 5 — one possible solution

```
import numpy as np

def preprocess_rts(raw_rts):
    valid_ms = np.array([r * 1000 for r in raw_rts if r is not None])
    trimmed = valid_ms[valid_ms < np.percentile(valid_ms, 95)]
    centered = trimmed - trimmed.mean()
    return centered
```



```
def run_experiment(participant_id, session_number,  
                  condition_label, stimulus_list,  
                  response_key_map, timeout_ms,  
                  practice_trials, fixation_duration_ms,  
                  feedback_enabled, log_file_path):  
  
    ...
```

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Refactoring direction: Group related parameters into a dataclass; pass that instead. ***

Exercise 6 — one possible solution (part 1)

```
from dataclasses import dataclass
```

```
@dataclass
```

```
class ExperimentConfig:
```

```
    condition_label: str
```

```
    stimulus_list: list
```

```
    response_key_map: dict
```

```
    timeout_ms: int
```

```
    practice_trials: int
```

```
    fixation_duration_ms: int
```

```
    feedback_enabled: bool
```

```
    log_file_path: str
```

Exercise 6 — one possible solution (part 2)

```
def run_experiment(participant_id, session_number,  
                   config: ExperimentConfig):  
    ...
```

Exercise 7

```
hits_a = sum(1 for t in block_a if t.correct)
total_a = len(block_a)
hr_a = hits_a / total_a

hits_b = sum(1 for t in block_b if t.correct)
total_b = len(block_b)
hr_b = hits_b / total_b
```


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Refactoring direction: Extract a reusable function; test it once. ***

Exercise 7 — one possible solution

```
def hit_rate(block):  
    return sum(1 for t in block if t.correct) / len(block)  
  
hr_a = hit_rate(block_a)  
hr_b = hit_rate(block_b)
```

Exercise 8

```
def integrand_uniform(self, p):  
    return self.likelihood(p) * self.uniform_prior(p)  
  
def integrand_beta(self, p):  
    return self.likelihood(p) * self.beta_prior(p)  
  
def integrand_jeffreys(self, p):  
    return self.likelihood(p) * self.jeffreys_prior(p)
```

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Refactoring direction: Extract the shared structure; pass the prior as an argument. ***

Exercise 8 — one possible solution

```
def integrand(self, prior, p):  
    return self.likelihood(p) * prior(p)  
  
# callers:  
# self.integrand(self.uniform_prior,    p)  
# self.integrand(self.beta_prior,      p)  
# self.integrand(self.jeffreys_prior,   p)
```

Exercise 9

```
def compute_median(trial_list):  
    trial_list.sort()  
    n = len(trial_list)  
    return trial_list[n // 2]
```

```
rts = [0.42, 0.31, 0.58, 0.29]
```

```
m = compute_median(rts)
```

```
print(rts)  # [0.29, 0.31, 0.42, 0.58] – caller's list was mutated
```

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Refactoring direction: Work on a local copy; leave the caller's data unchanged. ***

Exercise 9 — one possible solution

```
def compute_median(trial_list):  
    sorted_trials = sorted(trial_list)    # returns a new list  
    n = len(sorted_trials)  
    return sorted_trials[n // 2]  
  
rts = [0.42, 0.31, 0.58, 0.29]  
m    = compute_median(rts)  
print(rts)    # [0.42, 0.31, 0.58, 0.29] – unchanged
```

Exercise 10

```
class SignalDetection:

    # Set the d-prime value
    def set_dprime(self, value):
        self.dprime = value # store d-prime

    # Get the d-prime value
    def get_dprime(self):
        return self.dprime # return the threshold
```

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Refactoring direction: Remove comments that repeat the code; consider whether the accessors add value. ***

Exercise 10 — one possible solution

```
class SignalDetection:

    def __init__(self, dprime):
        self._dprime = dprime

    def set_dprime(self, value):
        # We don't want users to edit dprime per policy 22031.
        raise AttributeError("The dprime value cannot be edited.")

    def get_dprime(self):
        return self._dprime
```

```
summary = {  
    s: sum(t.rt for t in trials if t.subject==s)  
        / len([t for t in trials if t.subject==s])  
    for s in subjects  
}
```

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Refactoring direction: Extract a helper function; name the intermediate result. ***

Exercise 11 — one possible solution

```
def mean_rt(trials, subject):  
    subject_trials = [t for t in trials if t.subject == subject]  
    return sum(t.rt for t in subject_trials) / len(subject_trials)  
  
summary = {s: mean_rt(trials, s) for s in subjects}
```

```
from scipy.stats import norm

def analyse(sid, hit, far, mrt, srt, age, grp):
    if not (0 <= hit <= 1):
        raise ValueError("hit out of range")
    dp = norm.ppf(hit) - norm.ppf(far)
    return sid, dp, mrt / srt
```

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Refactoring direction: Group participant data into a dataclass; pass the object. ***

Exercise 12 — one possible solution (part 1)

```
from dataclasses import dataclass
```

```
@dataclass
```

```
class Participant:
```

```
    subject_id: str
```

```
    hit_rate: float
```

```
    false_alarm_rate: float
```

```
    mean_rt: float
```

```
    sd_rt: float
```

```
    age: int
```

```
    group: str
```


Exercise 12 — one possible solution (part 2)

```
from scipy.stats import norm

def analyse(p: Participant):
    if not (0 <= p.hit_rate <= 1):
        raise ValueError("hit_rate out of range")
    dp = norm.ppf(p.hit_rate) - norm.ppf(p.false_alarm_rate)
    return p.subject_id, dp, p.mean_rt / p.sd_rt
```